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ASR560X Series

Datasheet

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SIG BQB



About This Document

This document introduces the specifications of ASR560X: single-chip Bluetooth Low Energy 5.1 full feature (compatible with 5.2)/IEEE 802.15.4/2.4G Proprietary multi-protocol SoC solution, and it also supports BLE Mesh network.

Intended Readers

This document is mainly for engineers who use this chip to develop their own platform and products, for instance:

- PCB Hardware Development Engineer
- Software Engineer
- Technical Support Engineer

Included Chip Models

The product models corresponding to this document are as follows.

Model	Protocol	Core	SiP Flash	Function
ASR560X	BLE 5.1 full feature (compatible with 5.2) BLE SIG Mesh IEEE 802.15.4 2.4G Proprietary	ARM CM0+	1 MB/ 512 KB	AOA/AOD/Voice/IRTxRx/ Quadrature Decoder/Keypad/ 5V UART/5V GPIO/ Wi-Fi concurrent

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2023.07	V1.3.1	Updated Section 2.4.10 and Table 2-2.
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1. Overview

1.1 General

ASR560X is a family of highly integrated single-chip Bluetooth Low Energy 5.1 full feature (compatible with 5.2)/IEEE 802.15.4/2.4G Proprietary multi-protocol SoC solutions. Some models also support BLE Mesh Network (Relay/Proxy/Low power/Friend features). Detailed part number illustration is provided in Figure 1-1 and Table 1-1.

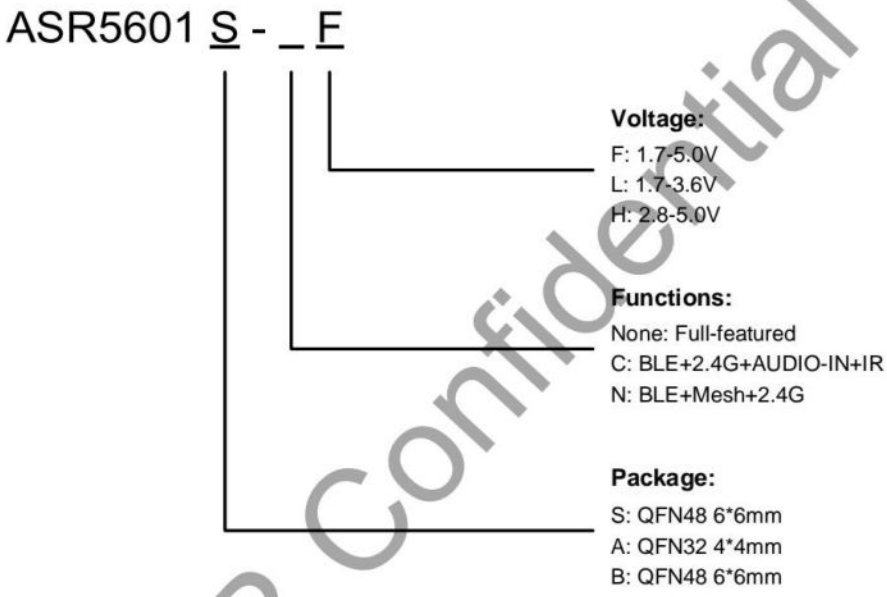


Figure 1-1 ASR560X Part Number

Table 1-1 ASR560X Part Number Information

Model	Sub	Protocol				AOA/	AUDIN/	Supply	Package	Flash	MCU	RAM
ASR560X	Code	BLE	Mesh	IEEE15.4	2.4G	AOD	IR	(V)				
ASR5601S	-F	√	√	√	√	√	√	1.7-5.0	QFN48	1MB	64MHz	96KB
ASR5601A	-NH	√	√		√			2.8-5.0	QFN32	512KB		
ASR5601A	-CL	√	√		√		√	1.7-3.6	QFN32			
ASR5601B	-CL	√	√		√		√	1.7-3.6	QFN48			

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Please note that all the following contents are mainly based on the full-featured model (ASR5601S-F). Some comments are added for differences of other models.

As illustrated below, ASR560X integrates all related blocks including the RF, the analog and the digital part. Additionally, the complete software package is available, which makes the development of user's own application software much easier.

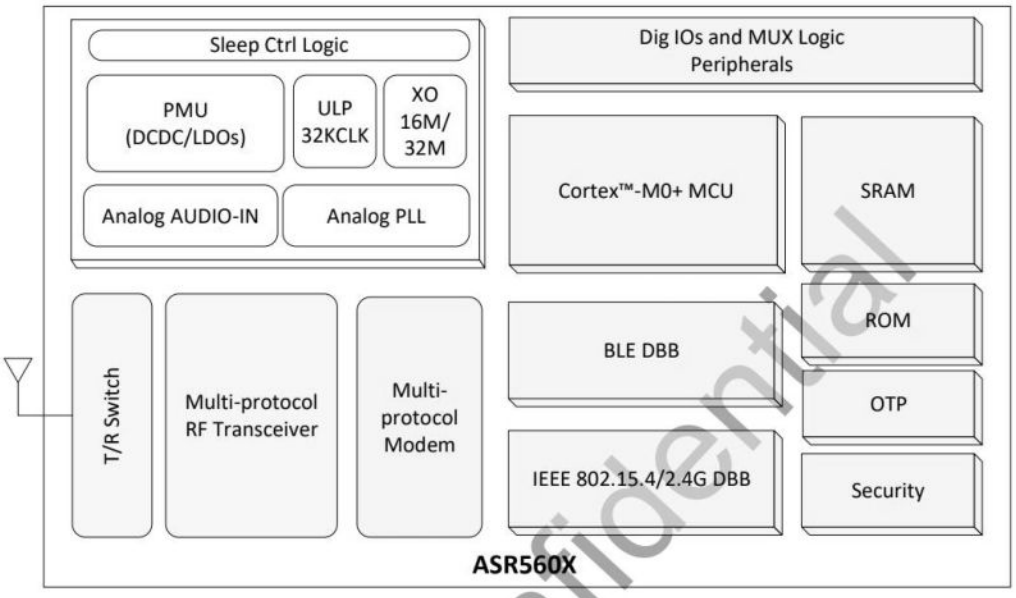


Figure 1-2 ASR560X Blocks Diagram

ASR560X has a multi-protocol RF transceiver and modem featuring high performance and low power consumption. It delivers a maximum of +10 dBm output RF power and achieves -98 dBm/-103.5 dBm sensitivity at BLE 1 Mbps/15.4 250 Kbps mode without off-chip power amplifier (PA) and TX/RX switch (TRSW). Moreover, an AUDIO-IN path supports both audio-in signals from analog and digital microphones (AMIC/DMIC). The analog audio-in path includes audio PGA, sigma-delta ADC (SD-ADC) as well as MIC bias voltage generation circuits (MICVB). This audio-in path enables flexible configuration to acquire audio signals for speech recognition in voice controlling application. The battery voltage range is extended from 1.7-3.6V, 2.8-5.0 V or 1.7-5.0 V with a flexible power management unit (PMU). The DCDC in PMU boosts the power efficiency of the entire chip. The ARM® Cortex™ M0+ MCU processes the multi-protocol's software stacks and user's application software. The Cortex™-M0+ core ensures high efficiency and low power. With multi-protocol RF and modem, this SoC supports data rates as 125K/500K/1M/2M bps in BLE mode and 250 Kbps in IEEE 802.15.4/Zigbee mode.

ASR560X is designed to meet various kinds of IoT applications such as the data transmission, the remote controller, the wearable devices, the speech recognition and AOA/AOD. OTP is integrated in order to enhance the system's security. A true random number generator (TRNG) is also integrated. Encryption of flash image file with key generated by TRNG protects user's software from illegal copying and ensures the safety of this SoC for IoT applications.

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1.2 Key Features and Performance

- **Certification**

  SIG BQB BLE 5.2 certified, Qualified Design ID (QDID): 174605

- **RF Transceiver and Modem**
  - Integrated PA and TRSW, single-ended RF Pin, simple interface to antenna
  - Two-point modulation transmitter with robust calibration algorithm
  - Low power receiver with configurable analog BB for 1 MHz and 2 MHz bandwidth
  - Multi-protocol RF transceiver and modem supporting Bluetooth Low Energy 5.1 full feature (compatible with 5.2), IEEE 802.15.4/Zigbee and 2.4G Proprietary
  - Support External PA
  - -95 dBm sensitivity at BLE 2 Mbps GFSK
  - -98 dBm sensitivity at BLE 1 Mbps GFSK
  - -103 dBm sensitivity at BLE 500 Kbps GFSK
  - -107 dBm sensitivity at BLE 125 Kbps GFSK
  - -103.5 dBm sensitivity at Zigbee OQPSK-DSSS for IEEE 802.15.4/Zigbee protocol
  - -30 dBm to +10 dBm output RF power with low harmonic emissions
  - Current under RX mode, 5.2 mA at 3 V VBAT with -98 dBm sensitivity 1 Mbps BLE
  - Current under TX mode, 4.7 mA at 3 V VBAT with 0 dBm output power
  - Current under TX mode, 18 mA at 3 V VBAT with +8 dBm output power
  - Various system sleep modes with less than 200 nA at ≤3.3 V VBAT in deep-sleep mode. When VBAT>3.6 V, sleep current will add about 10 uA due to HVLDO turning on.

  Note:

  For detailed information regarding external PA and sleep modes setting, please refer to ASR560X Application Notes.

- **Bluetooth Low Energy 5.1 full feature (compatible with 5.2)**
  - Bluetooth version 5.1 full feature (compatible with 5.2) specifications compliant, targeting BLE features only
  - BLE SIG Mesh: Relay/Proxy/Low power/Friend features
  - Support all packets types
  - Support all AOA/AOD features
  - Support WLAN coexistence mechanism (PTA)
  - Support all device roles (broadcaster, central, observer, peripheral)
  - Support encryption/decryption (AES-CCM) and bit stream processing (CRC,

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whitening)

- Support frequency hopping calculation (scheme #1 and scheme #2)
- Support FDMA/TDMA/events formatting and synchronization
- Support AHB slave interface for registers access and exchange memory of the CPU access via BLE core
- Support 16 MHz and 8 MHz operating clock
- Support power down BLE core during protocol's idle periods and runs at either 32.0 KHz or 32.768 KHz clock

- **IEEE 802.15.4/Zigbee**
  - IEEE 802.15.4 MAC specification 2011 version compliant
  - Support for both full function device (FFD) and reduced function device (RFD)
  - Support for both device and coordinator roles
  - Support non-beacon enabled mode, beacon enabled mode is not supported
  - Support all packets type (beacon/data/acknowledgment/MAC command)
  - Support FRAME encryption/decryption and authentication/authentication checking
  - Support promiscuous mode
- **2.4G Proprietary**
  - Support a star network topology with 1 Master node and up to 8 Slave nodes
  - Bidirectional data transfer with acknowledgement, retransmission and hardware CRC verification
  - Support CSMA-CA algorithm, which can avoid sending conflicts
  - Master node RX on idle, while Slave node open RX when needed, which can save power
  - Slave node can join or leave net freely
  - Support 40 channels with speed 250 Kbps
  - Channel hopping when current channel busy
  - Support security
- **Power Management Unit (PMU)**
  - Support single supply input pin, range from 1.7-3.6 V, 2.8-5.0 V or 1.7-5.0 V
  - Integrated DCDC (BUCK) and LDOs, no need for off-chip PMU
  - Integrated single-pin ultra-low power (ULP) 32.768 KHz XTAL circuit (XO32K)
  - ULP 32 KHz RC-Oscillator (RCO32K) with high accuracy and stability with auto-calibration
  - Finite status machine (FSM) under always-on power domain controls chip power status

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- **MCU System**
  - ARM Cortex™-M0+ MCU featuring low power and high efficiency
  - ARM Cortex™-M0+ with hardware multiplier
  - MCU can run at 16/64 MHz clock frequency configured by software
  - Boot ROM and 1K-bit OTP is integrated
  - 2x2K-Byte dedicated retention RAM in always-on domain
  - 96 KB system RAM configurable for Instruction or data
  - System RAM can be partially configured to retention/shut-down/sleep at low power mode
  - 1 MB or 512 KB on-system flash
  - Support execute-in-place (XiP) mode with on-system flash
  - Support external SPI NOR Flash
  - Support OTA (conditional)
- **Peripherals**
  - Support 5 V UART (UART0/UART1) and 5 V GPIO (P02/03/04/05)
  - Integrate PWM/TIMER/WATCHDOG/RTC
  - Integrate UART/SPI/I2C/DMA
  - Integrate GPIOs/KEYPAD/QUADRATURE DECODER
  - Integrate USB 2.0 full speed device
  - Integrate IR transceivers (IR0/IR1)
  - Integrate analog audio-in and digital I2S interfaces
  - Integrate temperature sensor and VBAT voltage detector
  - Integrate generic SAR ADC (AUX_ADC) with a maximum of 8 channels
- **Security**
  - Integrate true random number generator (TRNG)
  - Integrated OTP for key data storage and verification
  - Support flash image encryption/decryption
  - Support hardware dynamic encryption/decryption for customer data (AES-CCM)

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1.3 System Power Modes

There are five typical power modes in ASR560X, the ACTIVE, TX, RX, IDLE and deep sleep (DS-X). The SoC boots up and enters into the ACTIVE mode when it is powered on. The ACTIVE mode is the basic mode of this SoC. All other modes can only start from the ACTIVE mode and switch back to the ACTIVE mode. During the communication process, the soft stack requires the SoC to enter into the receiver (RX) or transmitter (TX) mode from the ACTIVE mode. If there is no TX or RX activity in a relatively short time, the chip will go to the IDLE mode to save power, in which the CPU enters into WFI mode and all the peripherals are suspended. If the spare time is long enough, the SoC will jump from the ACTIVE to the deep sleep (DS-X) mode to extremely save power consumption. It may be waked up by the GPIO events, internal time alarms or BLE anchor point requirement and then returns back to the ACTIVE mode.

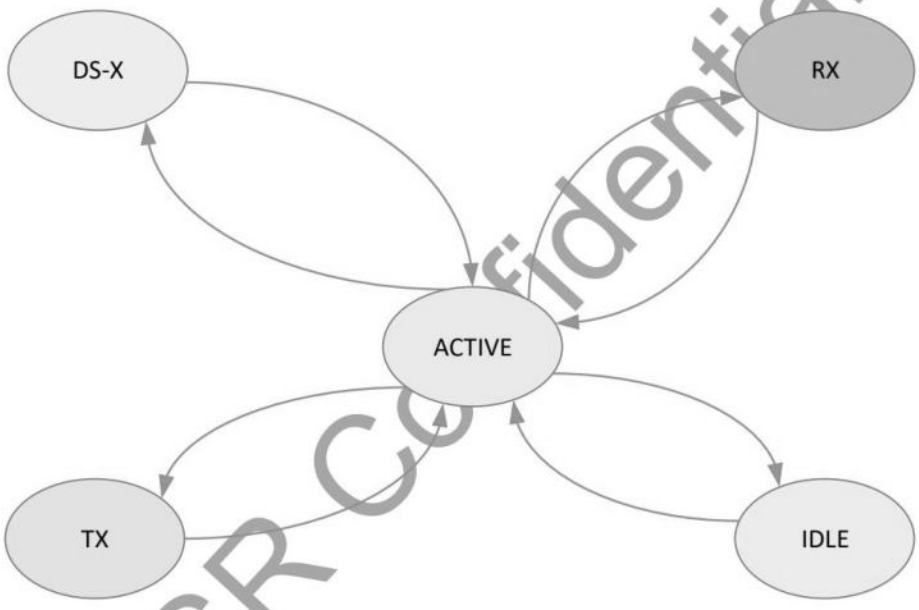


Figure 1-3 ASR560X Power Mode FSM

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Detailed sleep modes introductions are as follows:

- **DS1**

  DS1 mode supports GPIO wakeup. This mode cannot maintain BLE connection.

- **DS2**

  DS2 mode supports GPIO, RTC Timer, LPUART and BLE Timer wakeup. This mode cannot maintain BLE connection.

- **DS3**

  DS3 mode supports GPIO, RTC Timer, LPUART and BLE Timer wakeup, as well as beacon signal. This mode cannot maintain BLE connection.

- **DS4**

  DS4 mode supports GPIO, RTC Timer, LPUART and BLE Timer wakeup. This mode can maintain multiple BLE connections.

> **Note:**
> *For detailed information regarding sleep modes setting, please refer to ASR560X Application Notes.*

1.4 Applications

- Data Transmission
- BLE Voice Remote Controller
- Wearable Electronics
- Smart Plug and Lighting
- Smart Home
- BLE Keyboard and Mouse
- Positioning and Tag
- BLE Mesh Network
- Sensor Network

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2. MCU System

2.1 System Clocks

ASR560X supports 16M or 32M XTALs to generate the high-performance reference clock for RF/Analog PLLs. Two ULP 32.768 KHz clock sources are the one-pin ULP XO and the RC oscillator (RCO). They both work under the always-on power domain and the RCO features high stability and accuracy after auto-calibration. The low frequency clock for sleep mode can come either from RCO32K or XO32K. There are two analog PLLs. One generates the clocks for MCU and the peripherals. The other generates the clocks for the audio related circuits such as the audio-in sigma-delta ADC and the I2S interface. Both PLLs are referenced by the 16M/ 32M XO.

2.2 MCU Functions

2.2.1 Introduction

ASR560X integrates the ARM Cortex™-M0+ processor with internal RAM and ROM. The high-performance ARM Cortex-M0+ processor provides a low-cost platform which meets the needs of minimal memory implementation and low power consumption. It delivers outstanding computational performance and exceptional system response to interrupts.

2.2.2 Features

ARM Cortex™-M0+ Core Features:

- Tight integration of system peripherals reduces area and development costs
- Thumb instruction set combines high code density with 32-bit performance
- Support for single-cycle I/O access
- Power control optimization of system components
- Integrate sleep modes for low power consumption
- Fast code execution enables running the processor with a slower clock or increasing sleep mode time
- Optimized code fetching for reducing flash and ROM power consumption
- Hardware multiplier
- Deterministic, high-performance interrupt handling for time-critical applications
- Deterministic instruction cycle timing
- Support for system level debug authentication

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- Serial wire debug (SWD) reduces the number of pins required for debugging
- Support optional instruction trace

2.3 Memory

2.3.1 ROM

The internal ROM stores the boot image and is mainly used to program the flash image, perform secure boot or non-secure boot function according to the configuration of external mode selection pins.

2.3.2 SRAM

ASR560X family provides up to 96 KB on-chip SRAM. The internal RAM is used for not only the code and data memory but also BLE packet buffers and cache memory. It can be configured by software remap register. Internal RAM is capable of selective retention during power saving mode.

2.3.3 FLASH

ASR560X provides on-system QSPI flash devices with system-in-package (SiP) form. The hardware encryption and flash remapping functions protect developers' programs and data stored in on-system flash. The QSPI flash controller supports execute-in-place (XiP) mode with on-system flash.

2.3.4 OTP

ASR560X provides 1K-bit one-time programmable (OTP) memory. Hardware configuration parameters such as the flash-encryption parameters and customer-specific parameters such as the data encryption keys can be stored in OTP. These parameters are available immediately after the device resetting process finishes.

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2.3.5 Memory Mapping

The following table describes various MCU peripherals and how they are mapped to the processor memory.

Table 2-1 Memory Mapping Table

Modules	Start Address	End Address
FLASH	0x1000_0000	0x10FF_FFFF
SYS RAM	0x2000_0000	0x2001_7FFF
RET RAM0	0x2003_0000	0x2003_07FF
RET RAM1	0x2003_0800	0x2003_0FFF
UART0	0x4000_0000	0x4000_0FFF
UART1	0x4000_1000	0x4000_1FFF
UART2	0x4000_2000	0x4000_2FFF
SPI0	0x4000_3000	0x4000_3FFF
I2C0	0x4000_4000	0x4000_4FFF
DMA_CFG	0x4000_5000	0x4000_5FFF
Watchdog	0x4000_8000	0x4000_8FFF
I2S	0x4000_9000	0x4000_9FFF
TIMER	0x4001_1000	0x4001_1FFF
AUDIO_ADC	0x4001_3000	0x4001_3FFF
QUADRATURE DECODER	0x4001_4000	0x4001_4FFF
KEYPAD	0x4001_5000	0x4001_5FFF
SPI1	0x4001_7000	0x4001_7FFF
UART3	0x4001_8000	0x4001_8FFF
I2C1	0x4001_9000	0x4001_9FFF
AUX_ADC	0x4001_A000	0x4001_AFFF
UART4	0x4001_B000	0x4001_BFFF
PWM0	0x4002_0000	0x4002_0FFF
GPIO	0x4003_0000	0x4003_0FFF
OTP	0x4003_2000	0x4003_2FFF
BLE/15.4/2.4G	0x4010_0000	0x4010_FFFF
SYS_CON	0x4011_0000	0x4011_FFFF

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2.4 Peripherals

The following table gives a summary of all peripherals of ASR560X.

The details of each peripheral will be presented next.

Table 2-2 Peripherals Summary

Peripherals	Num.	Feature
GPIO	30 (48-Pin)/ 14 (32 Pin)	AHB peripheral
		Input/output configurable
		Rising/falling edge detection and interruption
UART	4	APB peripheral
		UART0 supports FIFO/DMA/CTS/RTS
		UART1 and 2 support FIFO/DMA
		UART3 supports base TX/RX
LPUART	1	LPUART supports 8 Bytes buffer
IR	1	APB peripheral
		2 channels
		Support NEC and user-defined protocols
SPI	2	APB peripheral
		SPI + TI SSP + NS micro wire
		Master or slave mode
I2C	2	APB peripheral
		Master or slave mode
		100 Kbps and 400 Kbps mode
I2S	1	APB peripheral
		Master or slave mode
		Full duplex, support TX and RX
		Audio data resolutions of 32 bits
USB	1	AHB peripheral
		USB 2.0 full speed device mode
		3 endpoints (except EP0)
PWM	12	APB peripheral
		Support 135.6 MHz charging mode
		16-bit timer
		Capture mode (only 6 channels can be configured as input capture function)
		Support 0%~100% PWM
TIMERS	8	4 hardware timers in PWM
		4 independent general hardware timers (supporting free-run mode and user-defined mode)
KEYPAD	1	APB peripheral
		Maximum 112 keys
QUADRATURE DECODER	1	APB peripheral
		3 independent timers for (X Y Z)
AUX_ADC	1	APB peripheral
		Software triggering mode or hardware auto mode
		Maximum 8 channels

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2.4.1 DMA

Direct memory access (DMA) provides high-speed data transfer between peripherals and memory or within the memory without the CPU participation. There are two DMA channels and sixteen handshakes with peripherals. The main features of DMA are as following.

- Compatible with AHB-Lite for the DMA transfers
- Compatible with APB for programming the registers
- A single AHB-Lite master for transferring data using a 32-bit address and a 32-bit data bus
- Configurable number of DMA channels, dedicated handshake/ programmable priority level for each channel
- Using fixed priority determined by the DMA channel number for each channel's priority level arbitrate
- Support three types of transferring: memory-to-memory, memory-to-peripheral and peripheral-to-memory
- Support multiple DMA cycle types and multiple DMA transfer data widths
- Each DMA channel can access a primary, and alternate, channel control data structure
- All the channel control data is stored in system memory in little-endian format
- Performs all DMA transfers using the single AHB-Lite burst type
- Destination data width is equal to the source data width
- Number of transfers in a single DMA cycle can be programmed from 1 to 1024
- Transferring address increment can be greater than the data width
- A single output to indicate when an ERROR condition occurs on the AHB bus

2.4.2 GPIO

General purpose input/output (GPIO) pins are fully configurable. They are multiplexed with other interfaces, such as I2C, SPI, UART and etc. The GPIO pins support the below features.

- IO voltage of P02/P03/P04/P05 follows VBATA, the other GPIOs follow VBATB (please note that when VBATA<3.3 V, all IO voltage follows VBATA)
- Fast GPIOs support single-CPU-cycle I/O operation
- Common GPIOs support two-CPU-cycle I/O operation
- Input mode: the input value can be read through register
- Output mode: the output value can be set through register
- Interrupt: the input can be set to edge-trigger or level-trigger to generate CPU interrupt
- Support 4 types of trigger: Low level, high level, falling edge and rising edge
- Internal pull-up or pull-down configurable

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2.4.3 UART

ASR560X provides four universal asynchronous receiver transmitter (UART) controllers. One of them supports auto-flow control. Features of UART are as following.

- TX/RX FIFO depth is 16 Bytes
- Programmable frame properties: number of data bits/frame (5~8), optional parity bit (odd/even), number of stop bits (1, 1.5, 2)
- Include additional FIFO status registers and shadow registers, which can be accessed by software
- Loopback mode for test
- Support DMA mode

2.4.4 LPUART

ASR560X provides one low power UART (LPUART). LPUART supports up to 9600 bps rate and can be used to wake up the chip from sleep mode.

2.4.5 IR

Two sets of completed analog IR transceivers are integrated to make ASR560X connected easily with the IR diode without external extra buffers. ASR560X supports 2 channels of infrared transmission and receiving. By programming the pulse waveform, the chip supports NEC and user-defined protocols.

2.4.6 SPI

ASR560X supports four-wire and full-duplex Motorola serial peripheral interface (SPI) protocol. There are two SPI, both can be configured as master or slave. They support two options of clock polarity (CPOL) and clock phase (CPHA). The main features of SPI are as following.

- Master or slave operation
- Programmable clock bit rate and prescaler
- Separated transmit and receive, first-in-first-out memory buffers, 16-bit width, 8 locations deep
- Programmable choice of interface operation, SPI, micro-wire or TI synchronous serial
- Programmable data frame sized from 4 to 16 bits
- Independent masking of transmit FIFO, receive FIFO and receive overrun interrupt
- Support direct memory access (DMA)
- Master SPI support up to 32 MHz, and the slave SPI supports up to 5 MHz

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2.4.7 I2C

The inter-integrated circuits (I2C) bus has two wires, the serial data (SDA) and the serial clock (SCL). It carries information between the devices connected to the bus. Each device is recognized by a unique address and can operate as either a transmitter or receiver, depending on the function of the device. The main features of I2C are as following.

- I2C unit is compliant to TWSI bus specification version 2.1 with the exception of support for the hardware general call,10-bit slave address
- Multi-master and arbitration support
- Support standard-mode operation up to 100 Kbps
- Support fast-mode operation up to 400 Kbps

2.4.8 I2S/AUDIO-IN Path

The audio signal sources can be the analog microphone (AMIC) or the digital microphone (DMIC). The AMIC's signal is digitized with integrated analog audio-in path as described in Section *3.6.3 AUDIO-IN Path*. The DMIC's signal is sent to I2S interface.

- I2S transmitter and/or receiver base on the Philip's I2S serial protocol
- Support only one channel for both transmitter and receiver
- Full duplex communication due to the independence of transmitter and receiver
- Master or slave mode of operation
- Audio data resolutions of 32 bits
- Support DMA mode

2.4.9 USB

The USB full speed device controller support USB 2.0 specification. The controller supports the full-speed (12 Mbps) and the low-speed (1.5 Mbps) transfer.

- Support full-speed (12 Mbps) signaling bit rate
- Support low-speed (1.5 Mbps) signaling bit rate
- Recover clock and data from USB
- Support 3 endpoints except EP0:
  - 3 unidirectional EPs
  - 1 unidirectional EP + 1 bidirectional EP
- 4 TX buffers each sized 32 Bytes, 1 RX buffer sized 32 Bytes

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2.4.10 PWM

Pulse width modulation (PWM) generates pulse sequences with programmable frequency and duty cycle for LCD, vibrators and other devices. The PWM provides totally 12 PWM channels and 6 of them can be configured as input capture function. The main features of PWM are as following.

- 12 PWM generators with 16-bit counter
- Pre-scaling divider by 1/2/4/8/16/32/64/128
- Up or down mode for PWM output
- Dead-band generator which can be bypassed
- 16-bit input capture (edge/time) mode
- 16-bit timer mode

2.4.11 Timer/Watchdog

Besides 4 timers in PWM, ASR560X has another 4 independent HW timers. Each timer has a 32-bit counter. The initial value of the counter ranging from 0x00000000 to 0xFFFFFFFF is set by software before the counter starts. The counter has two modes.

- **Free-running mode (default):**

  The timer loads the initial value and starts to count down. When the value reaches 0x0, it sends an interrupt to the CPU. Afterwards the counter restarts to count down from initial value to 0x0 and generates another interrupt to the CPU until it is disabled or reprogrammed.

- **User-defined mode:**

  The timer loads initial value and starts to count down. When the value reaches 0x0, it sends an interrupt to the CPU. Unlike the free-running mode, it then stops.

Additionally, there is a dedicated timer for the watch dog. When the system functions correctly, the watch dog timer generates an interrupt to inform the CPU to reset the watch dog timer. When a system malfunction occurs, the watch dog timer cannot be reset as normal. Then it generates a reset signal to reboot the system.

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2.4.12 Keypad

The Keypad controller supports:

- Up to 112 keys
- Manual or automatic scan
- Simultaneous operation of matrix keypad
- Continuous keypad polling
- Key de-bounce logic for the matrix keypad

2.4.13 Quadrature Decoder

The quadrature decoder provides buffered decoding of quadrature-encoded sensor signals:

- Decoding of digital waveform from off-chip quadrature encoder
- Optional input de-bounce filters
- 3 independent hardware timers for (X, Y, Z)

2.4.14 Temperature Sensor and VBAT Detector

A temperature sensor has been integrated for monitoring the temperature inside the chip and its output voltage is digitized by the 12-bit AUX_ADC. The resolution of the temperature sensor is around ±5 °C. A battery voltage detector is available for detecting the voltage of battery supply. Its output is also digitized by the 12-bit AUX_ADC. Details of the AUX_ADC can be found in Part AUX_ADC of Section *3.6.4 Miscellaneous Analog*.

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2.4.15 PIN MUX

All digital IOs can be reconfigured via software. The Pin Mux table is shown as follows. Note that 48-pin ASR560X has all 30 IOs, while 32-pin ASR560X has P00~P10 and P27~P29 IOs, of which P27, P28 and P29 can be used as GPIO or analog IOs for audio input.

Table 2-3 Digital Pin Mux Table -I

Num.	Pin Name	Func=0	Func=1	Func=2	Func=3	Func=4
1	P00	NA	UART2_TXD	I2C0_SCL	I2C1_SCL	PWM10
2	P01	NA	UART2_RXD	I2C0_SDA	I2C1_SDA	PWM11
3	P02	GPIO2	UART0_TXD	SPI0_CS	I2C0_SCL	PWM0
4	P03	GPIO3	UART0_RXD	SPI0_CLK	I2C0_SDA	PWM1
5	P04	GPIO4	UART1_TXD	SPI0_TXD	I2C1_SCL	PWM2
6	P05	GPIO5	UART1_RXD	SPI0_RXD	I2C1_SDA	PWM3
7	P06	SWC	UART3_TXD	SPI1_CS	I2S_SCLK	PWM4
8	P07	SWD	UART3_RXD	SPI1_CLK	I2S_LRCK	PWM5
9	P08	GPIO8	UART2_TXD	SPI1_TXD	I2S_DI	PWM6
10	P09	GPIO9	UART2_RXD	SPI1_RXD	I2S_MCLK	PWM7
11	P10	GPIO10	UART3_TXD	IR1	I2S_DO	PWM8
12	P11	GPIO11	UART1_TXD	SPI0_CS	I2C1_SCL	PWM9
13	P12	GPIO12	UART1_RXD	SPI0_CLK	I2C1_SDA	PWM10
14	P13	GPIO13	UART3_TXD	SPI0_TXD	I2C0_SCL	PWM11
15	P14	GPIO14	UART3_RXD	SPI0_RXD	I2C0_SDA	PWM0
16	P15	GPIO15	UART0_TXD	SPI1_CS	I2S_SCLK	PWM1
17	P16	GPIO16	UART0_RXD	SPI1_CLK	I2S_LRCK	PWM2
18	P17	GPIO17	UART0_CTS	SPI1_TXD	I2S_DI	PWM3
19	P18	GPIO18	UART0_RTS	SPI1_RXD	I2S_MCLK	PWM4
20	P19	GPIO19	UART2_TXD	SPI0_CS	I2C0_SCL	PWM5
21	P20	GPIO20	UART2_RXD	SPI0_CLK	I2C0_SDA	PWM6
22	P21	GPIO21	UART0_TXD	SPI0_TXD	I2C1_SCL	PWM7
23	P22	GPIO22	UART0_RXD	SPI0_RXD	I2C1_SDA	PWM8
24	P23	GPIO23	UART1_TXD	SPI1_CS	I2C0_SCL	PWM9
25	P24	GPIO24	UART1_RXD	SPI1_CLK	I2C0_SDA	PWM10
26	P25	GPIO25	UART3_TXD	SPI1_TXD	I2C1_SCL	PWM11
27	P26	GPIO26	UART3_RXD	SPI1_RXD	I2C1_SDA	PWM0
28	P27	GPIO27	UART1_TXD	UART2_RXD	I2C0_SCL	PWM1
29	P28	GPIO28	UART1_RXD	KEY_ROW4	I2C0_SDA	PWM2
30	P29	GPIO29	UART2_TXD	KEY_ROW5	I2S_DO	PWM3

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Table 2-4 Digital Pin Mux Table -II

Num.	Pin Name	Func=5	Func=6	Func=7	Func=8
1	P00	GPIO0	KEY_COL4	AXIS_2_P	NA
2	P01	GPIO1	KEY_COL5	AXIS_2_N	NA
3	P02	AXIS_0_P	KEY_ROW0	I2S_DI	SWC
4	P03	AXIS_0_N	KEY_ROW1	I2S_MCLK	SWD
5	P04	UART0_CTS	KEY_ROW2	LPUART_TXD[a]	I2C0_SCL
6	P05	UART0_RTS	KEY_ROW3	LPUART_TXD[a]	I2C0_SDA
7	P06	AXIS_1_P	KEY_COL0	LPUART_TXD[a]	GPIO6
8	P07	AXIS_1_N	KEY_COL1	LPUART_TXD[a]	GPIO7
9	P08	AXIS_2_P	KEY_COL2	USB_DP	NA
10	P09	AXIS_2_N	KEY_COL3	USB_DM	NA
11	P10	UART0_CTS	KEY_ROW4	NA	NA
12	P11	AXIS_1_N	KEY_ROW4	SWC	NA
13	P12	I2S_DO	KEY_ROW5	SWD	NA
14	P13	AXIS_0_P	KEY_COL4	LPUART_TXD	NA
15	P14	AXIS_0_N	KEY_COL5	LPUART_TXD	NA
16	P15	AXIS_1_P	KEY_ROW6	USB_DP	NA
17	P16	IR0	KEY_ROW7	USB_DM	NA
18	P17	AXIS_2_P	KEY_COL6	SWC	NA
19	P18	AXIS_2_N	KEY_COL7	SWD	NA
20	P19	AXIS_0_P	KEY_ROW8	LPUART_TXD	NA
21	P20	AXIS_0_N	KEY_ROW9	LPUART_TXD	NA
22	P21	AXIS_1_P	KEY_ROW10	NA	NA
23	P22	AXIS_1_N	KEY_ROW11	NA	NA
24	P23	AXIS_2_P	KEY_ROW12	LPUART_TXD	NA
25	P24	AXIS_2_N	KEY_ROW13	LPUART_TXD	NA
26	P25	NA	KEY_ROW2	NA	NA
27	P26	I2S_DO	KEY_ROW3	NA	NA
28	P27	KEY_COL0	KEY_ROW0	NA	NA
29	P28	KEY_COL1	KEY_ROW1	NA	NA
30	P29	KEY_COL2	KEY_ROW4	NA	NA

[a] If you need to use LPUART RXD, select one pad from P02~P26 through the configuration register, and configure the pad as GPIO (no input and no output mode). For more details, please refer to *2.4.3 UART* section.

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3. Sub-Blocks

ASR560X supports Bluetooth Low Energy 5.1 full feature (compatible with 5.2)/IEEE 802.15.4/ 2.4G Proprietary multi-protocol operation. It integrates a complete multi-mode RF transceiver and modem as well as PHY and MAC. An audio-in sub-system is also integrated which includes audio PGA and sigma-delta ADC for amplifying and digitizing signals from analog microphone (AMIC). Besides, digital microphone (DMIC) can be connected by I2S interface.

3.1 BLE Link Layer

- Bluetooth version 5.1 full feature (compatible with 5.2) specifications compliant, targeting BLE features only
- BLE SIG Mesh: Relay/Proxy/Low power/Friend features
- Support all packets types
- Support all AOA/AOD features
- Support WLAN coexistence mechanism (PTA)
- Support all device roles (broadcaster, central, observer, peripheral)
- Support encryption/decryption (AES-CCM) and bit stream processing (CRC, whitening)
- Support frequency hopping calculation (scheme #1 and scheme #2)
- Support FDMA/TDMA/events formatting and synchronization
- Support AHB slave interface for registers access and exchange memory of the CPU access via BLE core
- Support 16 MHz and 8 MHz operating clock
- Support power down BLE core during protocol's idle periods and runs at either 32.0 KHz or 32.768 KHz clock

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3.2 BLE PHY

- -95 dBm sensitivity at BLE 2 Mbps GFSK
- -98 dBm sensitivity at BLE 1 Mbps GFSK
- -103 dBm sensitivity at BLE 500 Kbps GFSK
- -107 dBm sensitivity at BLE 125 Kbps GFSK
- -30 dBm to +10 dBm output RF power with low harmonic emissions
- Slight receiver performance degradation with both large frequency offset and frequency drift impairments
- Robust support for BLE 125K/500K/1M/2M bps rates with large crystal ppm errors

- Robust support for BLE 125K/500K/1M/2M bps rates with large crystal ppm errors

3.3 IEEE 802.15.4/Zigbee MAC

- IEEE 802.15.4 MAC specification 2011 version compliant
- Support for both full function device (FFD) and reduced function device (RFD)
- Support for both device and coordinator roles
- Support non-beacon enabled mode, beacon enabled mode is not supported
- Support all packets type (beacon/data/acknowledgment/MAC command)
- Support FRAME encryption/decryption and authentication/authentication checking
- Support promiscuous mode

3.4 IEEE 802.15.4/Zigbee PHY

- -103.5 dBm sensitivity at IEEE 802.15.4/Zigbee 250 Kbps OQPSK-DSSS

3.5 2.4G Proprietary

- Support a star network topology with 1 Master node and up to 8 Slave nodes
- Bidirectional data transfer with acknowledgement, retransmission and hardware CRC verification
- Support CSMA-CA algorithm, which can avoid sending conflicts
- Master node RX on idle, while Slave node open RX when needed, which can save power
- Slave node can join or leave net freely
- Support 40 channels with speed 250 Kbps
- Channel hopping when current channel busy
- Support security

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3.6 RF and Analog

Besides the digital sub-systems, ASR560X integrates all necessary RF and analog sub-blocks to support the functions of the SoC as illustrated in Figure 3-1. The main RF and analog sub-blocks include a RF receiver, a RF transmitter, a RF PLL with two-point modulation, a power management unit (PMU) with always-on FSM, clock generation circuits (XO16M/32M, XO/RCO32K and APLL), a true random numbers generation (TRNG), a temperature sensor, an auxiliary ADC (AUX_ADC) and an analog micro-phone interface (AUDIO-IN).

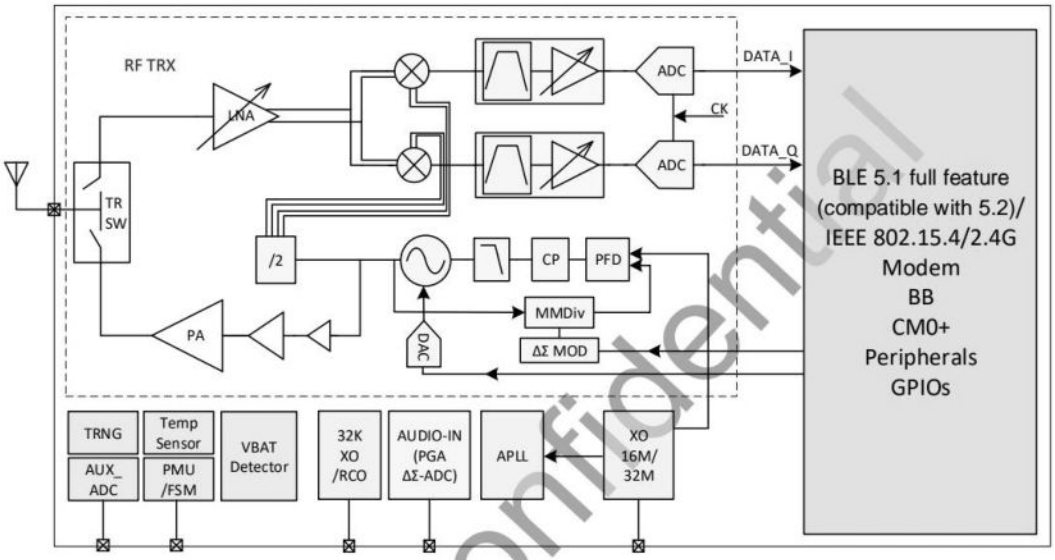


Figure 3-1 ASR560X RF and Analog Sub-Blocks

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3.6.1 RF Transceiver

The RF transceiver includes the receiver and transmitter as well as the RFPLL which generates the LO signals in receiver mode and modulate the BB signal in transmitter mode. With reconfiguration, the transceiver works well in BLE/IEEE 802.15.4/2.4G Proprietary mode. The TR switch (TRSW) and the differential to single-ended RF signal conversion transformer are integrated. The off-chip matching network can be very simple. The receiver features high sensitivity (-98 dBm at 1 Mbps BLE mode) and low power (5.2 mA at 3 V VBATA). The transmitter can deliver a maximum of +10 dBm and a minimum of -30 dBm RF output power at antenna port. The RFPLL includes a two-point modulation path for digital BB signals. With dedicated calibration for RFPLL, the transmitter modulation accuracy and stability are ensured. Following tables show the transmitter and receiver specifications respectively.

Table 3-1 BLE Transmitter Specification

Specification		Min.	Typ.	Max.	Unit
Maximum RF transmit power		8.5	+10	10.5	dBm
Peak power - Average power		0.34	0.35	0.38	dB
In-band emissions	-3 MHz	-47	-42	-40	dBm
	-2 MHz	-43	-40	-36	dBm
	+2 MHz	-43	-40	-36	dBm
	+3 MHz	-47	-42	-40	dBm
Modulation characteristics	F1(avg)	245	255	260	KHz
	F2(max) 99.9%	100	100	100	%
	F2(avg)/F1(avg)	0.85	0.88	0.9	
Center freq. deviation, F_n (n = 0,1,2,…k)		-1	10	15	KHz
Freq. drift, $\|F_0-F_n\|$ (n = 2,3, 4,…k)			6		KHz
Initial freq. drift, $\|F_1-F_0\|$			3		KHz
Max. freq. drift rate, $\|F_n-F_n-5\|$ (n = 6,7, 8,…k)			5		KHz/50μs
Harmonics (cable mode)		-40	-45	-50	dBm

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Table 3-2 BLE Receiver Specification - I

Parameter	Description	Min.	Typ.	Max.	Unit
Frequency Range		2402		2480	MHz
BLE 1 Mbps Rate	Sensitivity with dirty off	-98.5	-98	-96.5	dBm
	Sensitivity with dirty on	-96.5	-96	-95	dBm
	Maximum input level	0	10	11	dBm
	C/I Co-channel	4	4.5	5	dB
	C/I 1 MHz	-13	-11	-10	dB
	C/I 2 MHz	-36	-35	-33	dB
	C/I >2 MHz	-52	-46	-37	dB
	C/I Image	-37	-36	-30	dB
	C/I Image 1 MHz	-52	-49	-46	dB
BLE 2 Mbps Rate	Sensitivity with dirty off	-95.5	-95	-94.5	dBm
	Sensitivity with dirty on	-93.5	-93	-92.5	dBm
	Maximum input level	0	10	11	dBm
	C/I Co-channel	4	4.5	5	dB
	C/I 2 MHz	-13	-12	-11	dB
	C/I 4 MHz	-49	-47	-36	dB
	C/I >4 MHz		-52		dB
	C/I Image	-36	-35	-30	dB
	C/I Image 2 MHz		-52		dB
BLE 500 Kbps Rate	Sensitivity with dirty off	-103.5	-103	-102	dBm
	Sensitivity with dirty on	-102.5	-102	-101	dBm
	Maximum input level	0	10	11	dBm
	C/I Co-channel	3	3.5	5	dB
	C/I 1 MHz	-17	-16	-14	dB
	C/I 2 MHz	-37	-36	-34	dB
	C/I >2 MHz	-52	-51	-41	dB
	C/I Image	-41	-40	-36	dB
	C/I Image 1 MHz	-52	-51	-50	dB
BLE 125 Kbps Rate	Sensitivity with dirty off	-107.5	-107	-105	dBm
	Sensitivity with dirty on	-106	-105.5	-103.5	dBm
	Maximum input level	0	10	11	dBm
	C/I Co-channel	0.5	1	5	dB
	C/I 1 MHz	-20	-19	-18	dB
	C/I 2 MHz	-39	-38	-36	dB
	C/I >2 MHz	-52	-51	-41	dB
	C/I Image	-41	-40	-36	dB
	C/I Image 1 MHz	-52	-51	-50	dB

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Table 3-3 BLE Receiver Specifications -II

Parameter	Description	Min.	Typ.	Max.	Unit
Out-of-band Blocking	30 MHz-2000 MHz		-5		dBm
	2000 MHz-2400 MHz		-5		dBm
	2484 MHz-2999 MHz		-5		dBm
	3000 MHz-6000 MHz*		-5		dBm
Inter-modulation (Pblocker)	Psignal = -64 dBm Fblocker= +3 MHz/+6 MHz		-22.5		dBm
Input match S11		-15	-12	-8	dB
PER report integrity		100	100	100	%

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Notes:

1. *Dirty on or dirty off controls whether adding fixed frequency offset, frequency drift over time, modulation index inaccuracy and a low frequency modulation which is sinusoidal with deviation of 50 KHz and a modulation frequency of 1250 Hz to the ideal Bluetooth le signal.*
2. *Out-of-band blocking power from 3000 MHz-6000 MHz is -5 dBm with an exception of -31 dBm at 2×f channel.*

Table 3-4 IEEE 802.15.4 (250 Kbps) RX Performance (±500 KHz deviation)

Specification	Min.	Typ.	Max.	Unit	Comments
Sensitivity	-104	-103.5	-102	dBm	
Frequency Offset Tolerance	-150		+150	KHz	
Adjacent channel rejection (-1/+1 ch)		26/55		dB	Pwanted= -82 dBm
Adjacent channel rejection (-2/+2 ch)		49/64		dB	Pwanted= -82 dBm
Error vector magnitude (EVM)			2%		

Table 3-5 IEEE 802.15.4 (250 Kbps) TX performance (±500 KHz deviation)

Specification	Min.	Typ.	Max.	Unit	Comments
Output power, maximum setting	+8.9	+9.8	+10	dBm	
Output power, minimum setting	-36	-30	-27	dBm	
Programmable output	-30	0	+10	dB	+4 dBm ~ +10 dBm 2 dB/step -30 dBm ~ +3 dBm 6 dB/step

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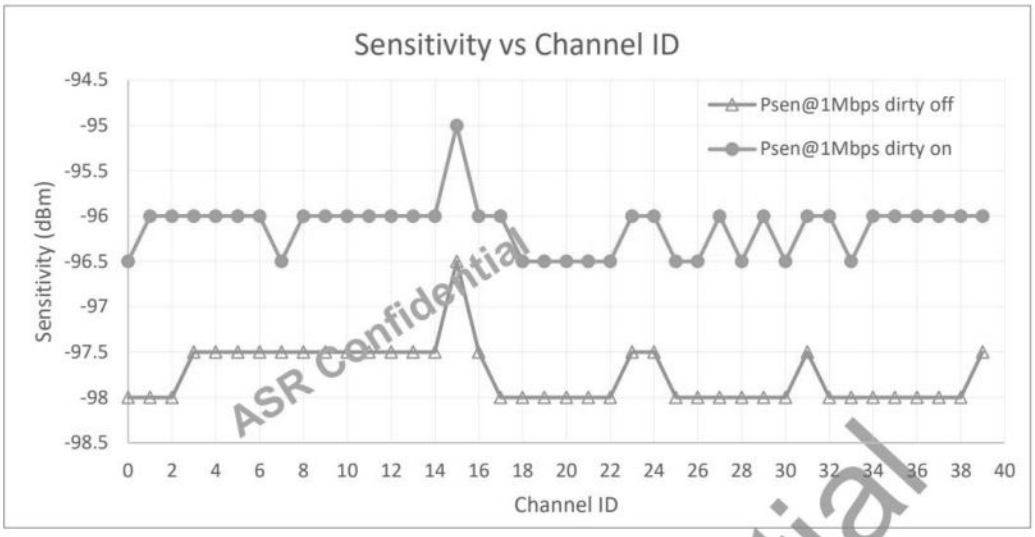


Figure 3-2 Sensitivity vs channel ID at BLE 1 Mbps

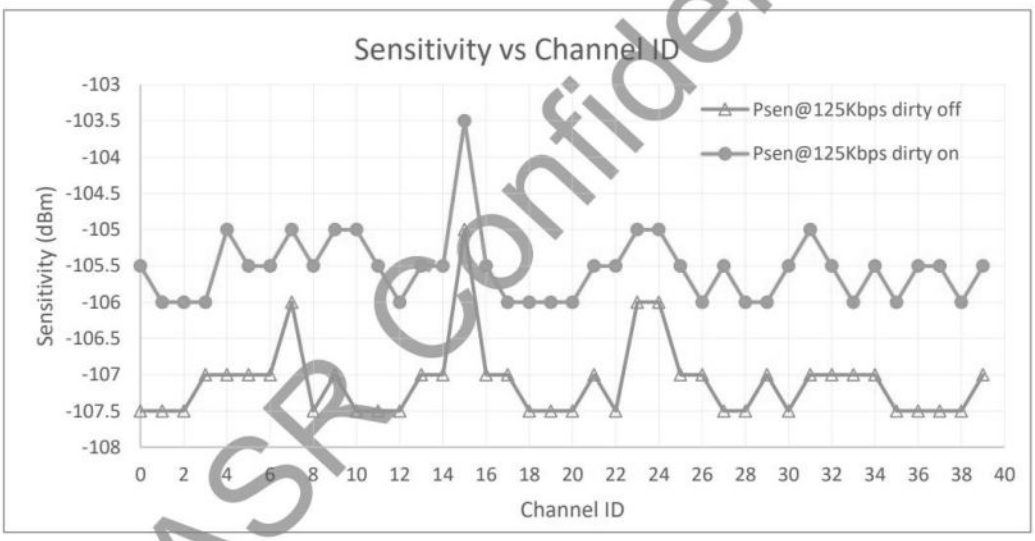


Figure 3-3 Sensitivity vs channel ID at BLE 125 Kbps

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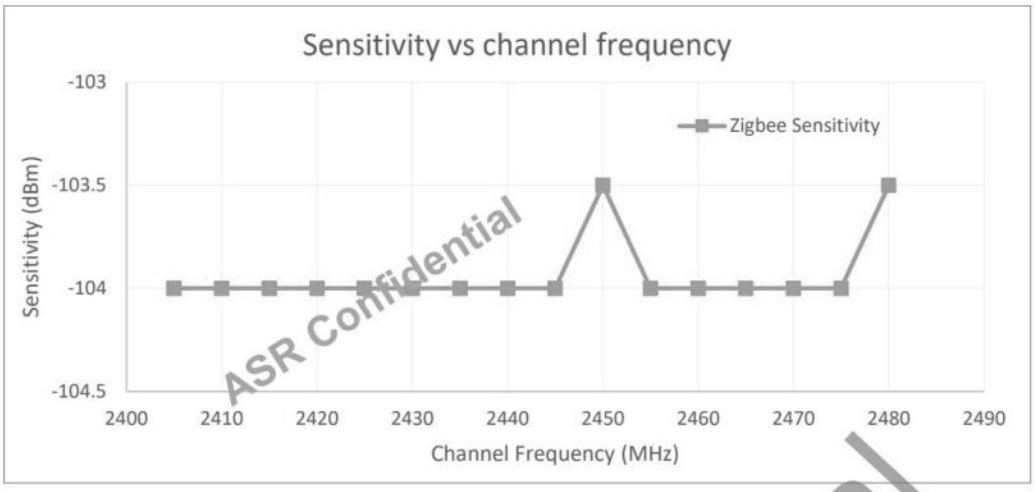


Figure 3-4 Sensitivity vs channel frequency at Zigbee 250 Kbps

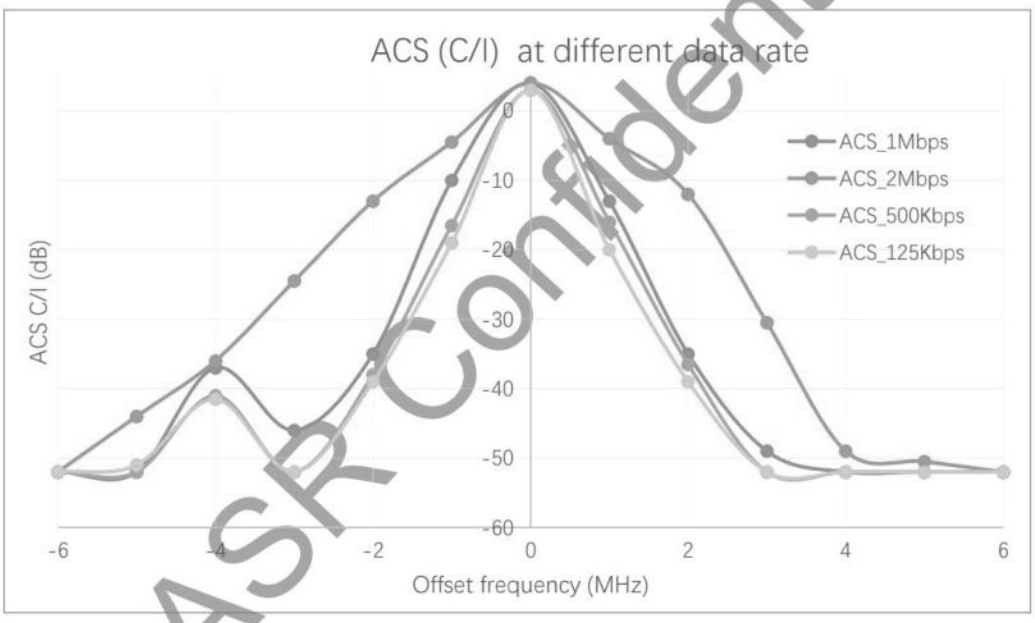


Figure 3-5 Adjacent channel selectivity at different data rate at BLE mode

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3.6.2 PMU

According to the part number details illustrated in Table 1-1, ASR560X can operate under 1.7-3.6 V, 2.8-5.0 V or 1.7-5.0 V supply voltage with single supply PIN. The PMU consists of a high voltage LDO, a DCDC (BUCK), a system LDO, a ULP self-regulated LDO, a ULP 32K (RCO and XO), a charger pump (CP), various LDOs for signal path and a system level always-on finite status machine (FSM). The high voltage LDO's outputs connected an off-chip capacitor at pin VBATB. The BUCK, which is paralleled with the system LDO. For low cost configuration the BUCK can be disabled while the system LDO powers the chip to save one bulky inductor. The CP is used to power the flash memory. Various LDOs with high PSRR and low noise are used to shield the noise-sensitive RF and analog circuits from the ripples of BUCK. The ULP self-regulated LDO powers the always-on FSM and the ULP 32K clock systems when the system enters into the deep sleep mode. The 32K clock system includes an RCO featuring high stability and accuracy after calibration and a nA-level XO 32K circuit. The always-on FSM controls the entire PMU to enter into various power modes according to the instruction from MCU to realize system level low power operation.

The BUCK consists of an LDO, a BGR, a soft-start block, a zero-current switching (ZCS) controller, a maximum current detection (MCD) controller, a logic controller, a driver, a power stage with 2 large MOSFETs and some off-chip passive devices. The key specification of the BUCK is shown in the table below.

Table 3-6 Performance Summary of BUCK

Specification	Min.	Typ.	Max.	Unit	Comments
VOUT	1.2	1.5	1.8	V	
Load Current	0	7.5	40	mA	
Efficiency		85.4	91.5	%	

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3.6.3 AUDIO-IN Path

The audio-in path is a high-performance analog to digital converter for differential microphone or line-in input. It consists of a programmable front-end filter (FE filter), a programmable gain amplifier (PGA), a continuous-time sigma-delta A/D converter and a microphone bias. The audio-in source can be either a microphone signal with a full-scale level as low as 16 mVppd or a line-in signal with up to 2 Vppd signal swing with different external ac-coupled capacitors (4.7 nF for MIC-in mode or 470 nF for LINE-in mode). The microphone bias provides the voltage varying from 1.5 V to 2.2 V (VBAT must 0.3 V higher than the output voltage) for the external microphone with at least 80 dB PSRR over the audio band (20 Hz to 22 KHz).

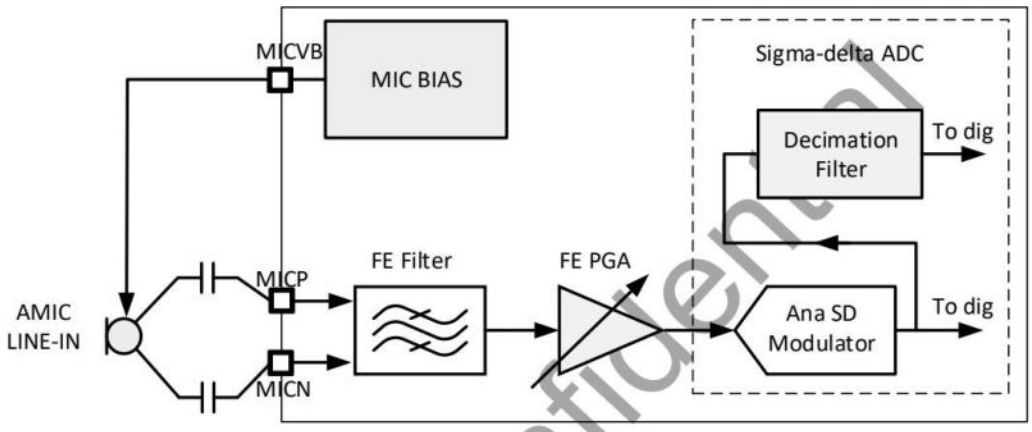


Figure 3-6 AUDIO-IN Path Diagram

Table 3-7 AUDIO-IN Path Performance Summary

Specification	Min.	Typ.	Max.	Unit	Comments
Filter Corner Frequency	100	1000	1000	Hz	100Hz/step
PGA Gain Range	24	42	42	dB	MIC-in, 3dB/step
	0	0	21	dB	LINE-in, 3dB/step
Dynamic Range	96	97	98	dB	0dB PGA gain, 22KHz BW
SNR	91.2	93.3	94.6	dB	0dB PGA gain, 0dBFS input
THD+N	-96.7	-96.3	-95.3	dBA	0dB PGA gain, 0dBFS input, A-weight
Input Referred Noise	14	16	20	μV	MIC-in, 42dB PGA gain, 22KHz BW
Data Rate	8/12/16/24/32/48/64/96/ 192/11.025/22.05/44.1/88.2			KHz	

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3.6.4 Miscellaneous Analog

- **AUX_ADC:**

  There are many other analog blocks to extend the usage of this SoC. The auxiliary ADC (AUX_ADC) is used to digitize low frequency analog signals with relative high-resolution requirements such as the output of on-chip temperature sensor and other external sensor, the battery voltage et al. It is based on SAR architecture with low cost, low power, flexibility and ease of use. It has a 12-bit simultaneous output with a maximum speed of 1 MS/s. The single-ended input voltage is buffered by a programmable gain amplifier (PGA) which has a gain range of 1x/1.5x/2x/2.5x/3x/4x/8x. The input range is from 0 V to 1.2 V. The AUX_ADC supports continuous and single-trigged conversion modes. Specially, it has a round-robin mode to automatically digitize all the channels specified with the orders configured in software. There are maximum 8 channels for external voltage digitization. The specification of the AUX_ADC is as following table.

Table 3-8 AUX_ADC Performance Summary

Specification	Min.	Typ.	Max.	Unit	Comments
Resolution		12		bits	
ENOB	10	10.5	11	bits	500 KHz input
Input Signal Bandwidth	DC		500	KHz	
Conversion Speed	0.125	1		MS/s	Configurable
No. of Channels			8		
Operation Mode	Continuous mode, single triggered mode and round-robin mode				

- **TRNG:**

  True random number generator (TRNG) generating random numbers is essential in security application. The TRNG uses random variation of the thermal noise voltage to obtain the 32-bit random data stream. It provides high-quality randomness because the chaotic system depends only on its initial condition rather than the statistics of the entropy source, hence resulting in a long-term unpredictability and insensitivity to the presence of deterministic noise.

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4. Software Package Specification

ASR560X software supports three modes, which are BLE 5.1 full feature (compatible with 5.2) mode, IEEE 802.15.4/Zigbee mode and 2.4G Proprietary mode. The software package provides bootloader, OTA and all peripheral drivers. It has a middleware APIs Layer, so that it can match different OSs, such as AliOS, FreeRTOS or NonOS.

4.1 BLE Mode

The BLE software provides BLE controller stack, host stack as well as mesh stack. In mesh operation, the software supports the PB-ADV, PB-GATT, provisioner and some mesh models. On the other hand, customers can also use the 3rd party's mesh models or develop their own models based on BLE mesh interface. In BLE mode, the software provides some common profiles as well as the stack. Also, customers can use the 3rd party's profiles or develop their own profiles.

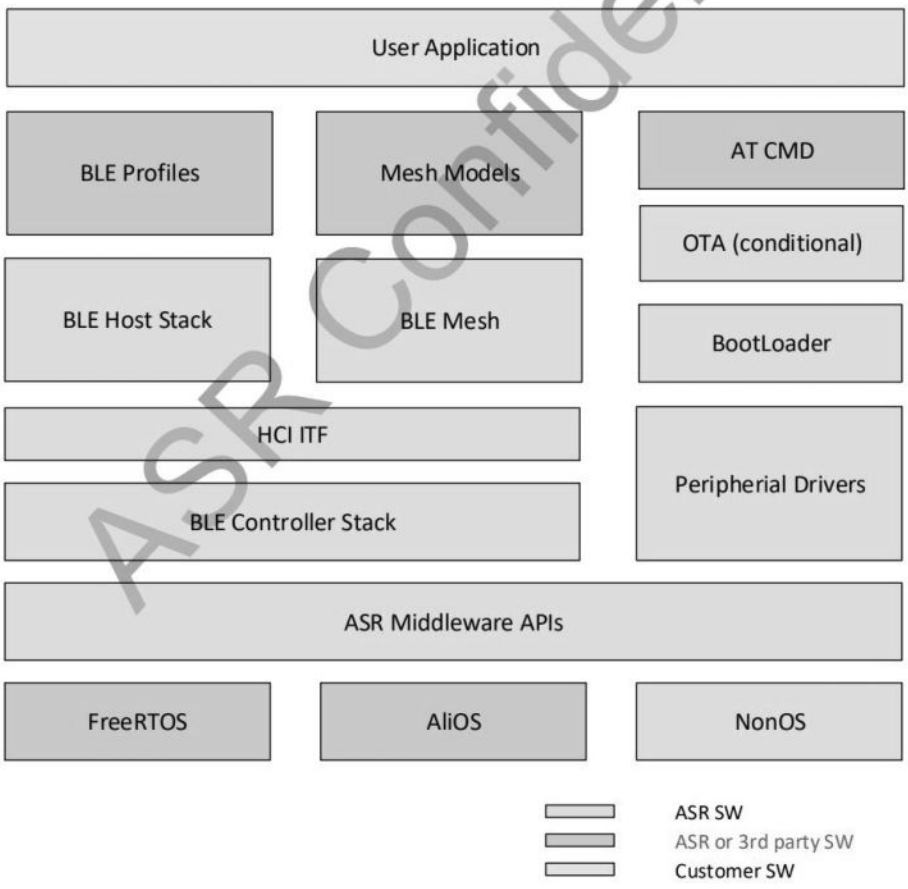


Figure 4-1 ASR560X BLE Software Architecture

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4.2 IEEE 802.15.4/Zigbee Mode

In Zigbee mode, the software package provides 802.15.4 stack and Zigbee stack. The OS/ Middleware APIs/peripheral drivers and OTA functions are the same as those in BLE mode. The Zigbee software package supports all roles, including coordinator, router and end-device.

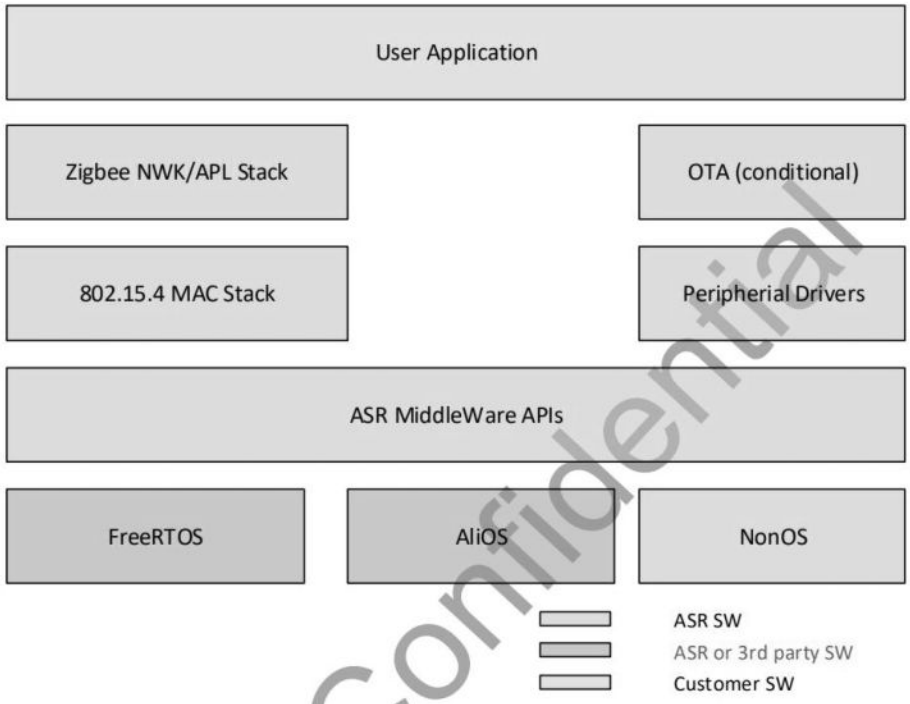


Figure 4-2 ASR560X Zigbee Software Architecture

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4.3 2.4G Proprietary Mode

The 2.4G proprietary software providers MACL stack and MAC stack. The MACL layer supports basic function, such as TX, RX, ACC, channel select etc. Customers can use this infrastructure to build protocol with more freedom. The MAC layer which based on MACL layer provider more function, such as pair, channel hopping, command control retransmission and security, more convenient to application.

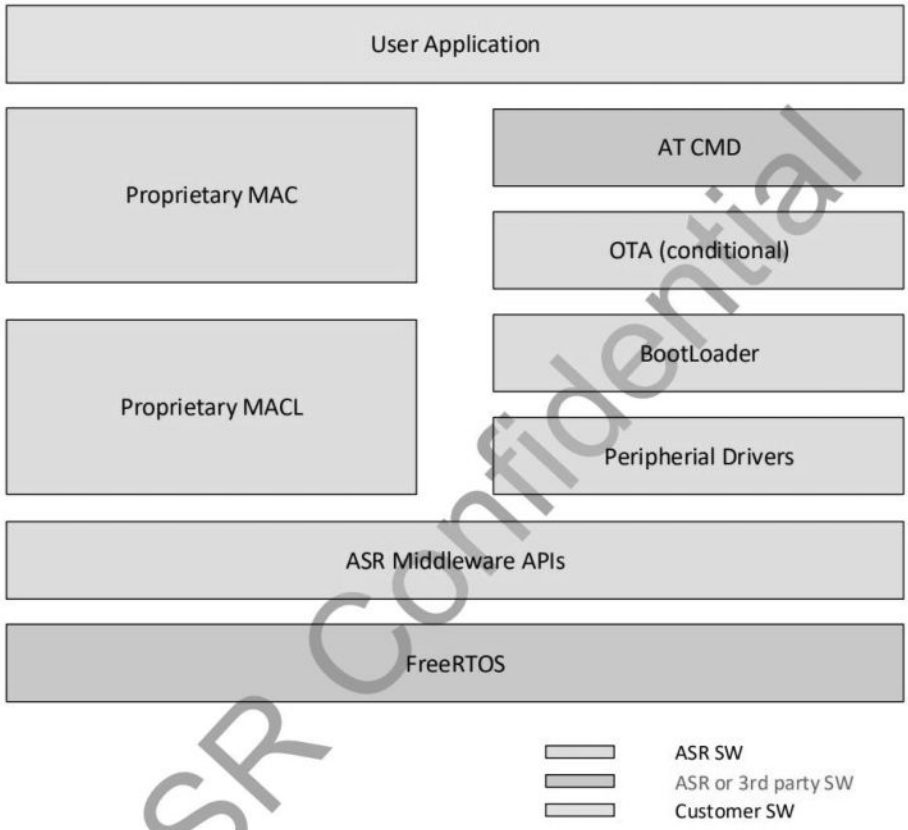


Figure 4-3 ASR560X 2.4G Proprietary Software Architecture

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5. Electrical Characteristics

5.1 Absolute Maximum Rating

Table 5-1 Absolute Maximum Rating

Specification	Symbol	Min.	Typ.	Max.	Unit
Power Supply	VBATA	-0.3		3.6	V
		-0.3		5.5	
		-0.3		5.5	
Storage Temperature Range	TSTR	-40		125	°C
Operating Temperature Range	TOPR	-20		85	°C

5.2 Recommended Operating Conditions

Table 5-2 Recommended Operating Conditions

Specification	Symbol	Min.	Typ.	Max.	Unit
Power Supply	VBATA	1.7	3.3	3.6	V
		2.8	4.2	5.5	
		1.7	5	5.5	
Operating Temperature Range	TOPR	-20	-	85	°C

Notes:

1. *ASR provides 560X SoC customization services which can support -20 to 105 °C or -20 to 125 °C operating temperature.*
2. *Power supply ranges are distinguished by different product name suffixes, please refer to Table 1-1 for more details.*

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5.3 Power Consumption in Sleep Mode

Table 5-3 Power Consumption of ASR560X QFN48 SoC under DS1-DS4

Sleep Mode	Current	Device	Note
DS1-A	0.2 uA	Tektronix DMM7510	T_A=25°C, V_{BAT}=3.3 V, Package: QFN48 Due to the very low current in Sleep mode, the data measured by different devices will vary slightly.
DS1-B	1.5 uA	Tektronix DMM7510	
DS2	1.7 uA	Tektronix DMM7510	
DS3	4.9 uA	Tektronix DMM7510	
DS4	15.7 uA	Tektronix DMM7510	

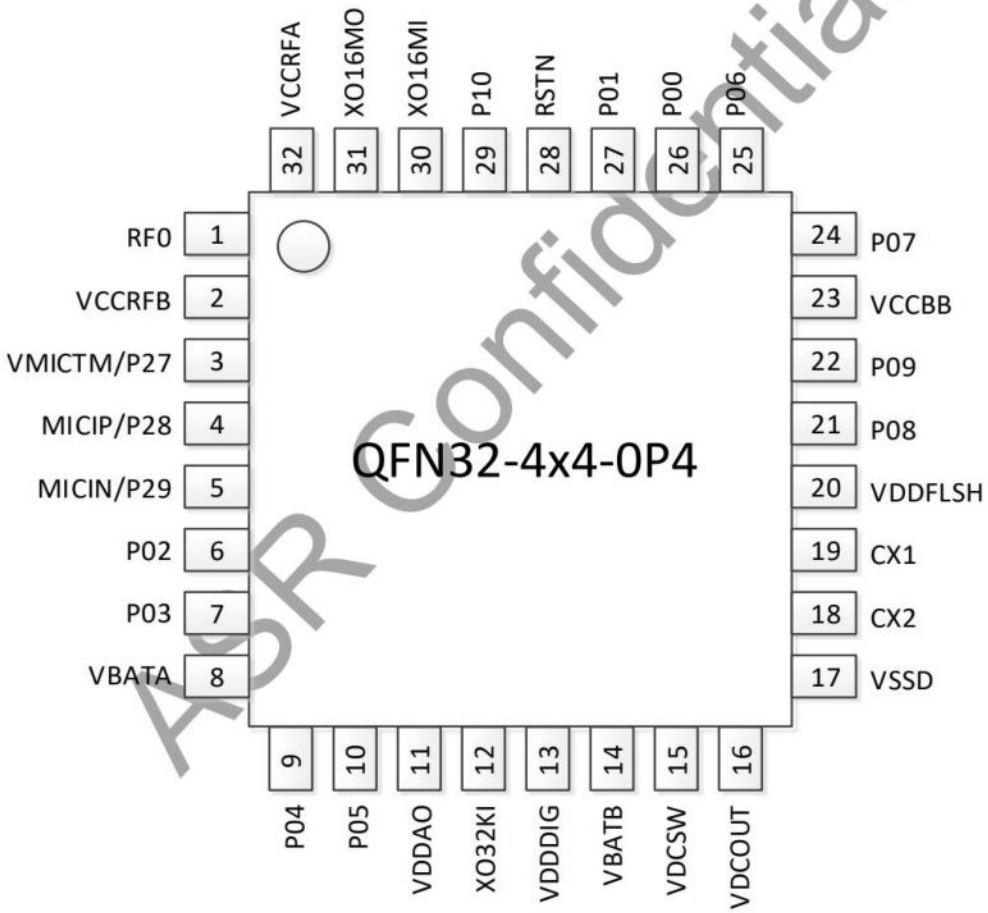
6. Package Specifications

With different numbers of digital IO available, there are two different packages for ASR560X, the 4 mm x 4 mm QFN package 32 pin and 6 mm x 6 mm QFN package 48 pin.

Both package forms are with 0.4 mm pitch.

6.1 QFN32

6.1.1 QFN32 Pin Map



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Figure 6-1 Pin Mapping of QFN32

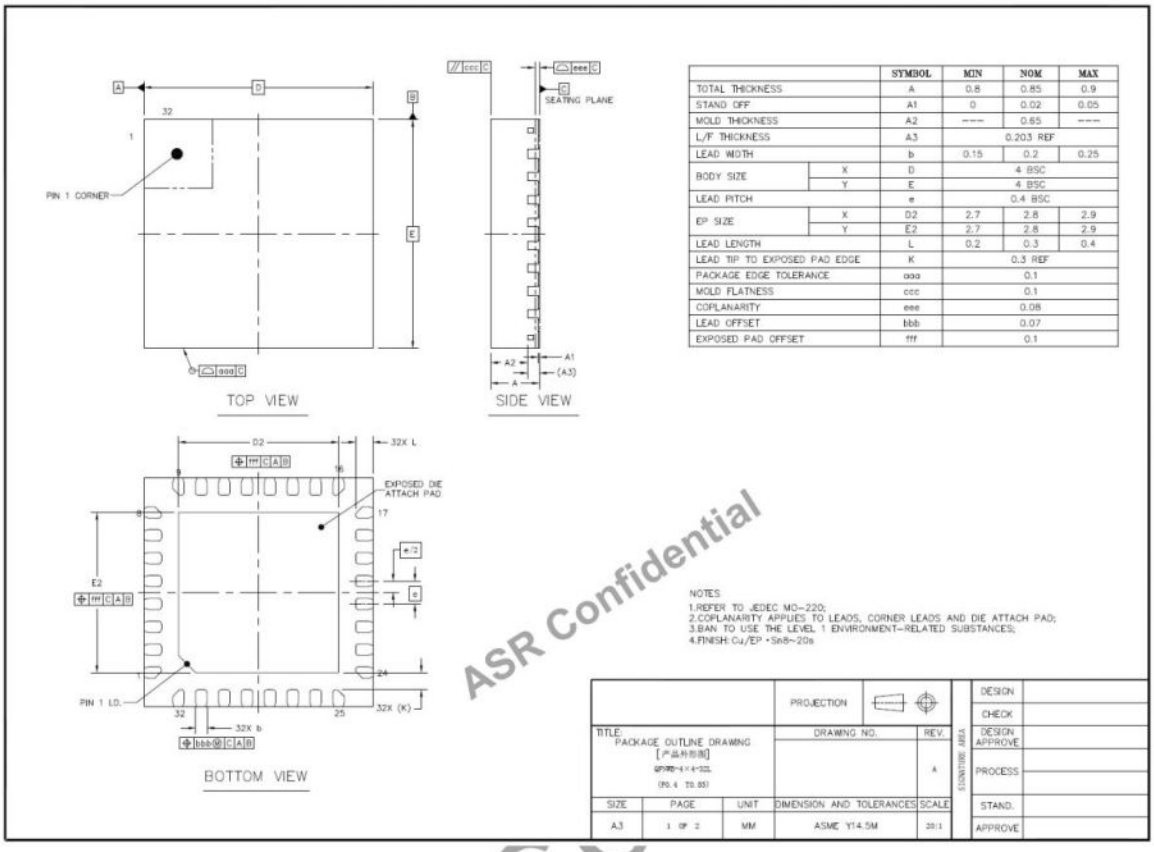
6.1.2 QFN32 Pin Description

Table 6-1 Pin Description of QFN32

Num.	Name	Property	Pin Description
1	RF0	RF	RF input and output
2	VCCRFB	Power	RF power supply input B
3	VMICTM/P27	Analog	Microphone bias voltage
4	MICIP/P28	Analog	Microphone/line input positive terminal
5	MICIN/P29	Analog	Microphone/line input negative terminal
6	P02	Digital I/O	See PIN MUX table
7	P03	Digital I/O	See PIN MUX table
8	VBATA	Power	System power supply, decap 1 μF to gnd
9	P04	Digital I/O	See PIN MUX table
10	P05	Digital I/O	See PIN MUX table
11	VDDAO	Power	Internal LDO output, decap 1 μF to gnd
12	XO32KI	Analog	XO 32K input
13	VDDDIG	Power	Internal LDO output, decap 1 μF to gnd
14	VBATB	Power	Decap 1 μF to gnd
15	VDCSW	Analog	DCDC switch, 4.7 μH inductor to DCDCOUT
16	VDCOUT	Power	DCDC output, 4.7 μH inductor to VDCSW, 2.2 μF to gnd
17	VSSD	Gnd	Digital ground
18	CX2	Analog	CP flying cap terminal 2, 100 nF to CX1
19	CX1	Analog	CP flying cap terminal 1, 100 nF to CX2
20	VDDFLSH	Power	CP output (Flash power supply), decap 1 μF to gnd
21	P08	Digital I/O	See PIN MUX table
22	P09	Digital I/O	See PIN MUX table
23	VCCBB	Power	Analog BB power input
24	P07	Digital I/O	See PIN MUX table
25	P06	Digital I/O	See PIN MUX table
26	P00	Digital I/O	See PIN MUX table
27	P01	Digital I/O	See PIN MUX table
28	RSTN	Analog	Chip reset, reset low, active floating
29	P10	Digital I/O	See PIN MUX table
30	XO16MI	Analog	XO 16M/32M input
31	XO16MO	Analog	XO 16M/32M output
32	VCCRFA	Power	RF power supply input A

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6.1.3 QFN32 Mechanical Parameters



		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.8	0.85	0.9
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	----	0.65	----
L/F THICKNESS		A3		0.203 REF	
LEAD WIDTH		b	0.15	0.2	0.25
BODY SIZE	X	D		4 BSC	
	Y	E		4 BSC	
LEAD PITCH		e		0.4 BSC	
EP SIZE	X	D2	2.7	2.8	2.9
	Y	E2	2.7	2.8	2.9
LEAD LENGTH		L	0.2	0.3	0.4
LEAD TIP TO EXPOSED PAD EDGE		K		0.3 REF	
PACKAGE EDGE TOLERANCE		aaa		0.1	
MOLD FLATNESS		ccc		0.1	
COPLANARITY		eee		0.08	
LEAD OFFSET		bbb		0.07	
EXPOSED PAD OFFSET		fff		0.1	

Figure 6-2 Package Outline Drawing of QFN32

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6.2 QFN48

6.2.1 QFN48 Pin Map

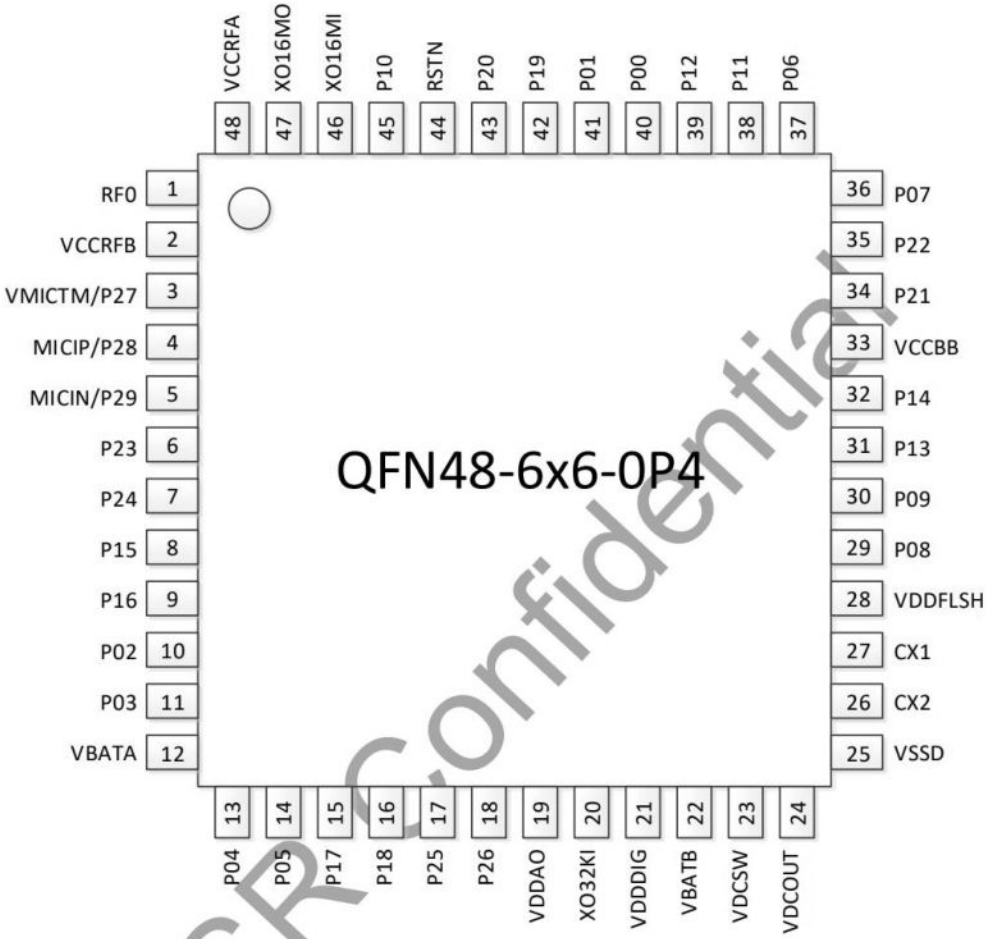


Figure 6-3 Pin Mapping of QFN48

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6.2.2 QFN48 Pin Description

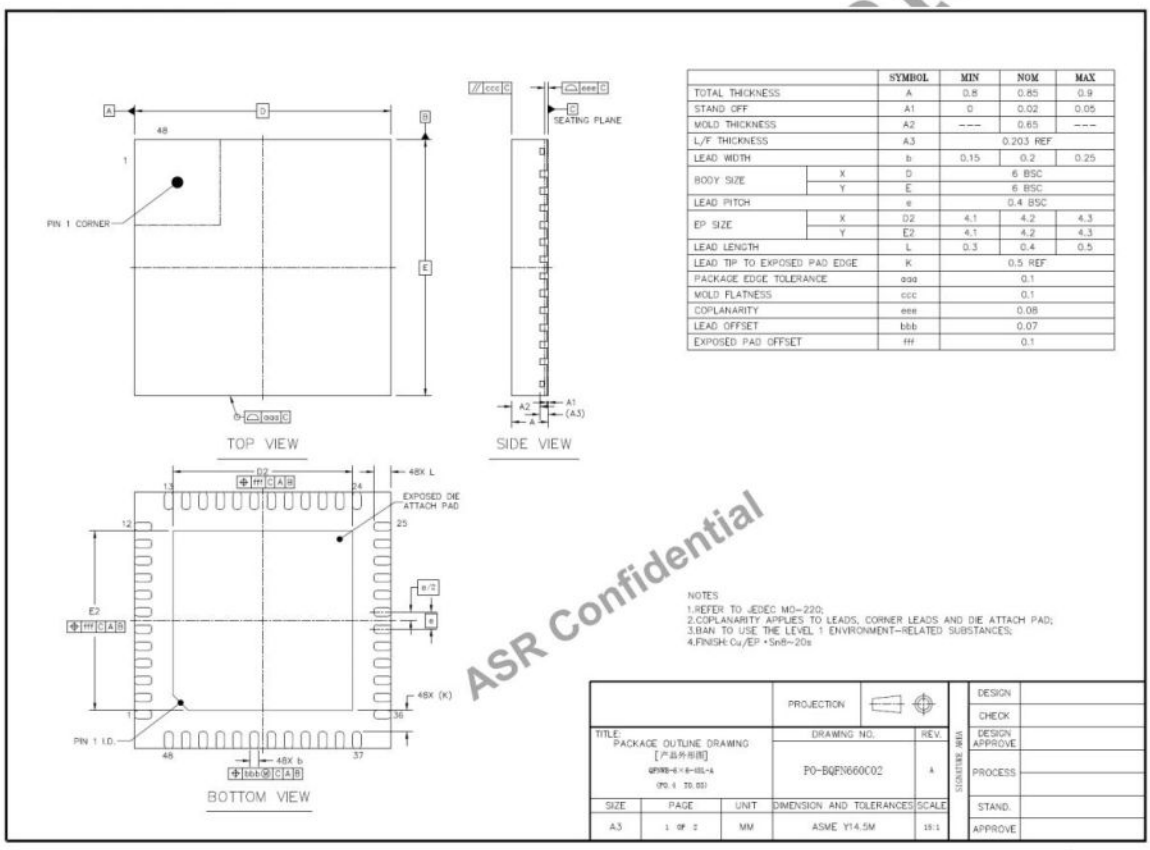
Table 6-2 Pin Description of QFN48

Num.	Name	Property	Pin Description
1	RF0	RF	RF input and output
2	VCCRFB	Power	RF power supply input B
3	VMICTM/P27	Analog	Microphone bias voltage
4	MICIP/P28	Analog	Microphone/line input positive terminal
5	MICIN/P29	Analog	Microphone/line input negative terminal
6	P23	Digital I/O	See PIN MUX table
7	P24	Digital I/O	See PIN MUX table
8	P15	Digital I/O	See PIN MUX table
9	P16	Digital I/O	See PIN MUX table
10	P02	Digital I/O	See PIN MUX table
11	P03	Digital I/O	See PIN MUX table
12	VBATA	Power	System power supply, decap 1 μF to gnd
13	P04	Digital I/O	See PIN MUX table
14	P05	Digital I/O	See PIN MUX table
15	P17	Digital I/O	See PIN MUX table
16	P18	Digital I/O	See PIN MUX table
17	P25	Digital I/O	See PIN MUX table
18	P26	Digital I/O	See PIN MUX table
19	VDDAO	Power	Internal LDO output, 1 μF-decap to gnd
20	XO32KI	Analog	XO 32K input
21	VDDDIG	Power	Internal LDO output, 1 μF-decap to gnd
22	VBATB	Power	Decap 1 μF to gnd
23	VDCSW	Analog	DCDC switch, 4.7 μH inductor to DCDCOUT
24	VDCOUT	Power	DCDC output, 4.7 μH inductor to VDCSW, 2.2 μF to gnd
25	VSSD	Gnd	Digital ground
26	CX2	Analog	CP flying cap terminal 2, 100 nF to CX1
27	CX1	Analog	CP flying cap terminal 1, 100 nF to CX2
28	VDDFLSH	Power	CP output (Flash power supply), 1 μF-decap to gnd
29	P08	Digital I/O	See PIN MUX table
30	P09	Digital I/O	See PIN MUX table
31	P13	Digital I/O	See PIN MUX table
32	P14	Digital I/O	See PIN MUX table
33	VCCBB	Power	Analog BB power input
34	P21	Digital I/O	See PIN MUX table
35	P22	Digital I/O	See PIN MUX table
36	P07	Digital I/O	See PIN MUX table
37	P06	Digital I/O	See PIN MUX table
38	P11	Digital I/O	See PIN MUX table

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Num.	Name	Property	Pin Description
39	P12	Digital I/O	See PIN MUX table
40	P00	Digital I/O	See PIN MUX table
41	P01	Digital I/O	See PIN MUX table
42	P19	Digital I/O	See PIN MUX table
43	P20	Digital I/O	See PIN MUX table
44	RSTN	Analog	Chip reset, reset low, active floating
45	P10	Digital I/O	See PIN MUX table
46	XO16MI	Analog	XO 16M/32M input
47	XO16MO	Analog	XO 16M/32M output
48	VCCRFA	Power	RF power supply input A

6.2.3 QFN48 Mechanical Parameters



		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.8	0.85	0.9
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.65	---
L/F THICKNESS		A3		0.203 REF	
LEAD WIDTH		b	0.15	0.2	0.25
BODY SIZE	X	D		6 BSC	
	Y	E		6 BSC	
LEAD PITCH		e		0.4 BSC	
EP SIZE	X	D2	4.1	4.2	4.3
	Y	E2	4.1	4.2	4.3
LEAD LENGTH		L	0.3	0.4	0.5
LEAD TIP TO EXPOSED PAD EDGE		K		0.5 REF	
PACKAGE EDGE TOLERANCE		aaa		0.1	
MOLD FLATNESS		ccc		0.1	
COPLANARITY		eee		0.08	
LEAD OFFSET		bbb		0.07	
EXPOSED PAD OFFSET		fff		0.1	

			PROJECTION		DESIGN
TITLE: PACKAGE OUTLINE DRAWING [产品外形图] QFN48-6*6-0.4-L-A (P0.4 T0.65)			DRAWING NO.	REV.	CHECK
			P0-BQFN660C02	A	DESIGN APPROVE
					PROCESS
SIZE	PAGE	UNIT	DIMENSION AND TOLERANCES	SCALE	STAND.
A3	1 OF 2	MM	ASME Y14.5M	10:1	APPROVE

Figure 6-4 Package Outline Drawing of QFN48

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